Huynh Doan Minh Ngoc

PhD Student

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Education

- **Sungkyunkwan University, South Korea**
  PhD Student in Computer Engineering *2023 – Present*
- **University of Science, Vietnam National University in Ho Chi Minh city**
  Bachelor in Computer Science, High-Quality Program *2017 – 2021*
  GPA: **3.82/4.00** (Valedictorian of the High-Quality Program)

Publications

- **Huynh, N. D. M.**, Tran, D. N. N., Pham, L. H.,... & Jeon, J. W. (2026). "TSBOW: Traffic Surveillance Benchmark for Occluded Vehicles Under Various Weather Conditions". In: *Proceedings of the AAAI Conference on Artificial Intelligence, 40(7), 5239–5247.* https://doi.org/10.1609/aaai.v40i7.37439 Github
- Le, T., **Huynh, N.**, Le, B. "Knowledge graph embedding by projection and rotation on hyperplanes for link prediction". In: *Appl Intell* (2022). https://doi.org/10.1007/s10489-022-03983-6
- Le, T., **Huynh, N.**, Le, B. (2021). "RotatHS: Rotation Embedding on the Hyperplane with Soft Constraints for Link Prediction on Knowledge Graph". In: *Nguyen, N.T., Iliadis, L., Maglogiannis, I., Trawiński, B. (eds) Computational Collective Intelligence. ICCCI 2021. Lecture Notes in Computer Science(), vol 12876. Springer, Cham.* https://doi.org/10.1007/978-3-030-88081-1_3
- Le, T., **Huynh, N.**, Le, B. (2021). "Link Prediction on Knowledge Graph by Rotation Embedding on the Hyperplane in the Complex Vector Space". In: *Farkaš, I., Masulli, P., Otte, S., Wermter, S. (eds) Artificial Neural Networks and Machine Learning – ICANN 2021. ICANN 2021. Lecture Notes in Computer Science(), vol 12893. Springer, Cham.* https://doi.org/10.1007/978-3-030-86365-4_14 Github

Workshops

- Tran, D. N. N., Pham, L. H.,... & **Huynh, N. D. M.**,... & Jeon, J. W. (2026). "A Hybrid Data-Centric Framework for Thermal Multiple-Object Tracking with Complex Motion Patterns". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 7200-7209).* Tran_CVPRW_2026_paper.html
- Tran, T. H. P., Tran, D. N. N., **Huynh, N. D. M.**, Tran, C. D.,... & Jeon, J. W. (2025). "DepthTrack: Cluster Meets BEV for Multi-Camera Multi-Target 3D Tracking". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 5289-5298).* Tran_ICCVW_2025_paper.html Github
- Pham, L. H., Ho, Q. P. N., Vu, D. K.,... & **Huynh, N. D. M.**,... & Jeon, J. W. (2025). "Data augmentation is all you need for robust fisheye object detection". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 5334-5342).* Pham_ICCVW_2025_paper.html
- Pham, L. H., Ho, Q. P. N.,... & **Huynh, N. D. M.**,... & Jeon, J. W. (2024). "Improving object detection to fisheye cameras with open-vocabulary pseudo-label approach". In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 7100-7107). Pham_CVPRW_2024_paper.html Github
- Nguyen, H. H., Tran, C. D.,... & **Huynh, N. D. M.**,... & Jeon, J. W. (2024). "Multi-View Spatial-Temporal Learning for Understanding Unusual Behaviors in Untrimmed Naturalistic Driving Videos". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 7144-7152).* Nguyen_CVPRW_2024_paper.html Github

Working Experience

- **Automation Lab, Sungkyunkwan University, South Korea** February 2023 - Present
  Assistant Researcher
- **FPT Software, Ho Chi Minh City, Viet Nam** February 2022 - February 2023
  Data Engineer

Projects

- **Intelligent Traffic Surveillance Benchmark**: Investigate the Object Detection task; Propose the TSBOW dataset,capturing occluded vehicles under various weather conditions
  Programming language: Python
- **Link Prediction on the Knowledge Graph based on Rotation Embedding method (Undergraduate Thesis)**: Investigate the Geometric approach in the Link Prediction problem; Give the proposed model (RotatHS) to improving the performance; Conduct some experiments with the RotatHS model
  Programming language: Python
- **Visualize and analyse the situation of COVID-19 in Vietnam**: Crawl Covid-19 data from Worldometer and Ministry of Health; Visualize data using the Tableau application; Write report and preparing slides; Apply the ARIMA and OLS models to predict the cases in the following day using *sklearn* and *statsmodels* libraries
  Programming language: Python
- **Credit card fraud detection**: Visualize data using the *matplotlib* library; Apply some models to classify the fraudulent transactions; Write report and preparing slides
  Programming language: Python
- **Student management application**: Design and code the UI; Create database; Code and Test the features
  Programming languages: C# (for UI) and SQL (for database)

Activities

- **The 40th Annual AAAI Conference on Artificial Intelligence** — Jan 20th - 27th, 2026
  Participate as the first author of accepted paper
  TSBOW: Traffic Surveillance Benchmark for Occluded Vehicles Under Various Weather Conditions
- **Female Developer Innovation Tournament 2021** — Dec 01st, 2021 - Feb 26th, 2022
  Champion and Best Innovation Idea with Lumière team — *The detailed idea*
- **13th International Conference on Computational Collective Intelligence** — Sep 28th - Oct 1st, 2021
  Participate as an author of accepted paper
  RotatHS: Rotation Embedding on the Hyperplane with Soft Constraints for Link Prediction on Knowledge Graph
- **30th International Conference on Artificial Neural Networks 2021** — Sep 18th - 17th, 2021
  Participate as an author of accepted paper
  Link Prediction on Knowledge Graph by Rotation Embedding on the Hyperplane in the Complex Vector Space
- **SheCodes Hackathon in HCMC** — Jul 18th - 19th, 2020
  First Place with VGO team

Honor and Awards

- **Graduated with Excellent degree classification for the academic year 2017-2021** — Apr 14th, 2022
  Vietnam National University, Ho Chi Minh City
- **Graduated as Valedictorian of the High-Quality Program 2017-2021** — Apr 12th, 2022
  University of Science, VNU-HCMC
- **Excellent thesis topic** — Nov 26th, 2021
  The Science and Technology Research Student Award 2021

Skills

- **Languages**: English (**IELTS**: 5.5)
- **Computer**: Latex
- **Programming Language**: Python, C++
- **Used to use**: C#, SQL, Java, Cuda C/C++

Soft Skills

- Critical thinker with excellent problem solving skills
- Adapt quickly to new environments
- Good interpersonal and communication skills

Objective

- Working and dedicating in a professional environment, respect each other, friendly,...
- Programming skills are trained in 4 years, capable of management (with the completion of a good team leader in most university projects) or jobs requiring a high level of technical expertise
- Research skills are trained during the course of the thesis (with 1 publication in Q2 journal and 3 accepted papers in 3 international conferences)